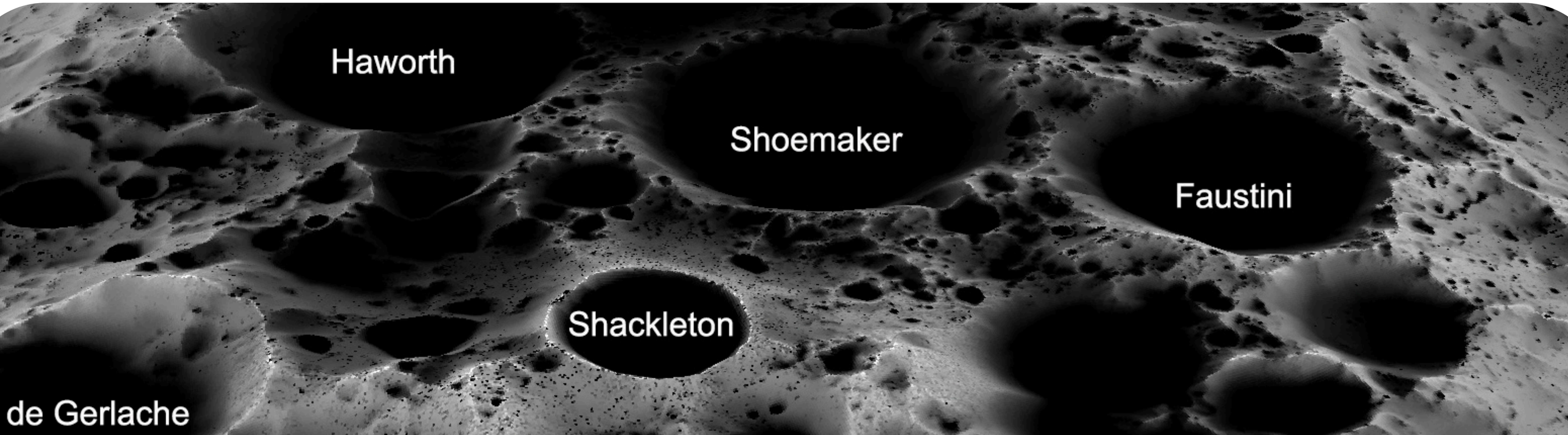
AITHON 2.0

TEAM NAME : TEAM SHAURYAM

TEAM LEADER NAME : PANKAJ JADHAV

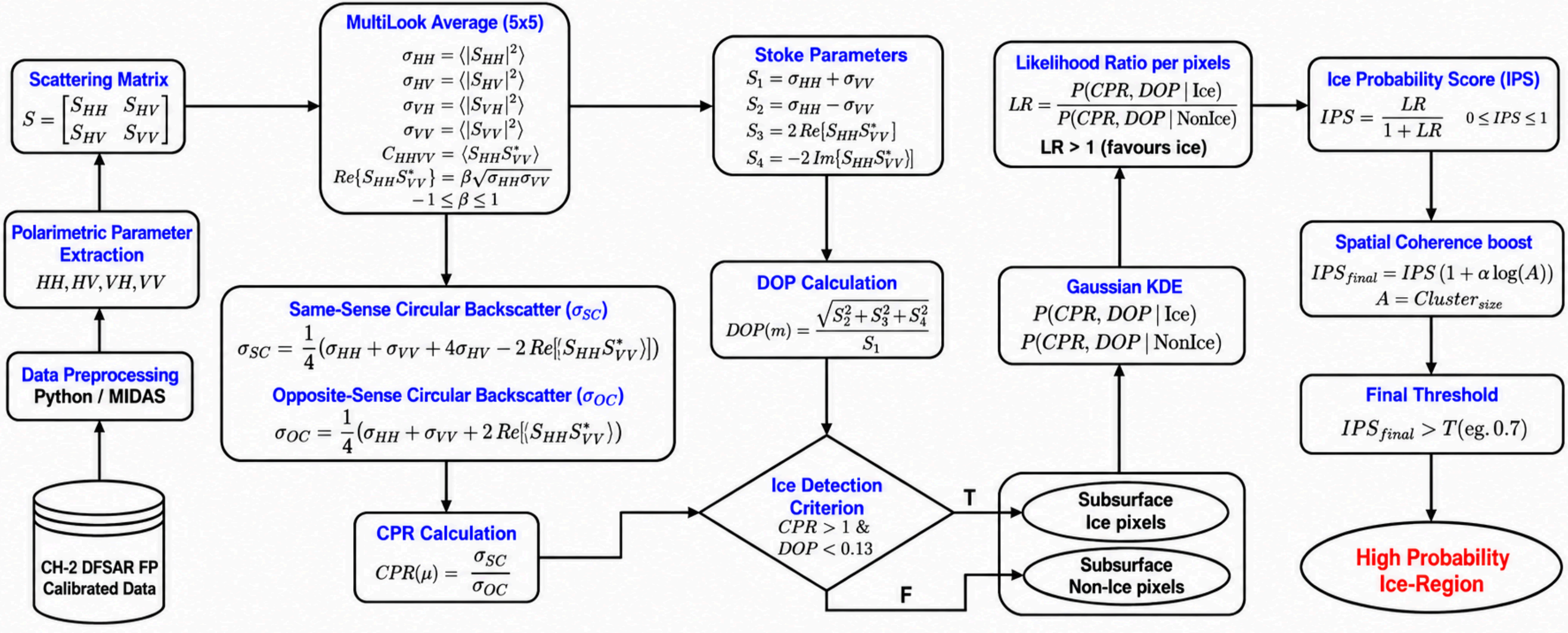
PS CATEGORY: SOFTWARE



IDEA TITLE

PROJECT / IDEA NAME:-	**Detection & Characterization of Subsurface Ice in Lunar South Polar Regions Using Chandrayaan-2 DFSAR Radar + OHRC Imagery for Landing Site Selection & Rover Traverse Planning**
TAGLINE:-	**From radar echoes to rover tracks: finding lunar water, and the safest way to reach it.**
PROBLEM BEING ADDRESSED:-	**Water ice at the Moon's south pole is vital for future missions, but it is buried under permanently shadowed craters. Today, ice detection, landing site selection, rover navigation, and resource estimation are treated as separate problems, so there is no single pipeline that takes us from raw radar data to a mission-ready plan.**
PROPOSED SOLUTION:-	**An end-to-end AI pipeline using Chandrayaan-2 DFSAR and OHRC data that goes from ice detection to usable ice estimates:** **Ice detection: CPR, DOP, and a KDE likelihood ratio give a pixel-wise Ice Probability Score.** **Landing site: A Fuzzy Cognitive Map ranks safe 1×1 km sites near ice.** **Rover path: DIC3D-A* plans the illuminated approach and PPO-DRL handles the battery-gated crater descent.** **Ice mass: An IEM-trained ANN ensemble and LLL mixing estimate ice volume and mass with uncertainty.** **Bonus: An ISRU workflow turns the ice into water, oxygen, and fuel.** **Key differentiator: one unified, uncertainty-aware pipeline from radar data to landing site, rover route, and ice mass.**

Objective 1: Identification of high-probability subsurface ice regions



AIESA
ARTIFICIAL INTELLIGENCE & DATA SCIENCE ENGINEERING STUDENT ASSOCIATION

TEAM SHAURYAM

AITHON 2.0

Objective 2: Feasible Landing Sites

1) Determination of sampling regions

- OHRC → Map PSRs & Doubly Shadowed Craters
- CH-2 DFSAR FP → Identify potential Subsurface ice Signature
- CPR > 1 & DOP < 0.13
- Slope (LOLA) → Proposed flat regions in ice reach regions for sampling
- Selected High Probability Ice-reach regions

2) Preselection of potential landing regions for the lander

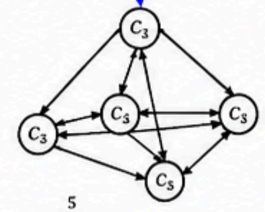
- Slope (LOLA) ≤ 8°
- DFSAR (block 1) ice-positive pixel
- Average illumination (LOLA) ≥ 35 %
- Generation of 10 x 10 m² seeds of landing cells and removal of artifact pixels
- Sliding Window RMS → Proposed 1 x 1 km² safe landing cells
- Ice detected Distance ≤ 15 km → Landing cells nearby selected ice-reach regions
- Rank → Potential landing regions → Selected landing regions

3. Characterization of selected landing regions and identification of optimal sites

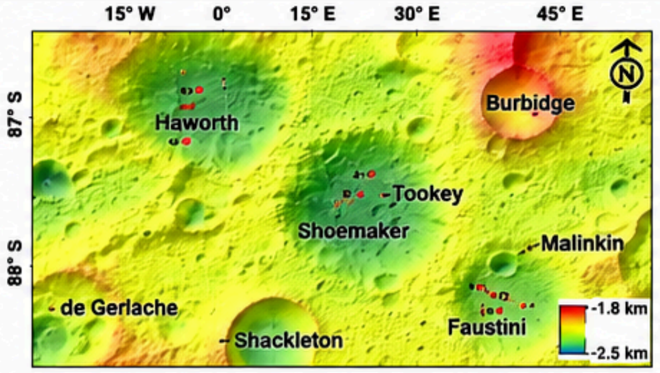
Identification of small-scale hazards: small craters	Identification of small-scale hazards: rocky region	High-resolution illumination condition analysis
OHRC images	DFSAR	LOLA DEM
Distribution of Shadow-casting small craters	Distribution of rocky regions	High-resolution illumination, duration of day and night

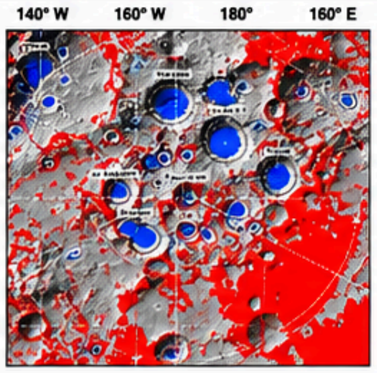
4) FCM-Based Optimal Landing Site Ranking

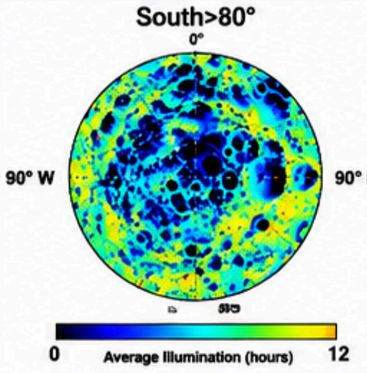
- Distance to Ice-region (C_1)
- Rocks distribution (C_3)
- Slope (C_2)
- Illumination intensity (C_4)
- Maximum temperature (C_5)
- Fuzzy Membership Functions
  - $\mu_i(x, v) = \min(\mu_i(x), \mu_i(v))$
  - $f(x, v) = \frac{2\,x\,y}{1 + x\,y}$
- Fuzzy Cognitive Map

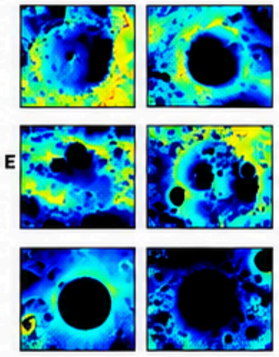


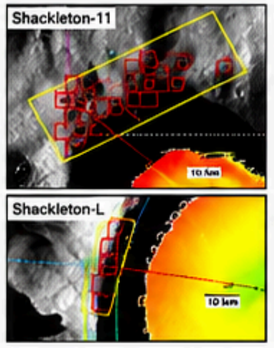
- Mamdani Minimum Fuzzy Rule Base
- Rule Aggregations (Family Einstein Sum)
- FCM State Update (Kosko method)
- $A_i^{t+1} = f\left(\sum_{j=1}^{5} A_j^t W_{ji}\right)$
- Evaluation Index Score (Final)
- $\text{Score} = \sum_{j=1}^{5} \omega_j y_j$
- $\sum_{j=1}^{5} \omega_j = 1,\ y_j \in [0,1]$
- Landing Site

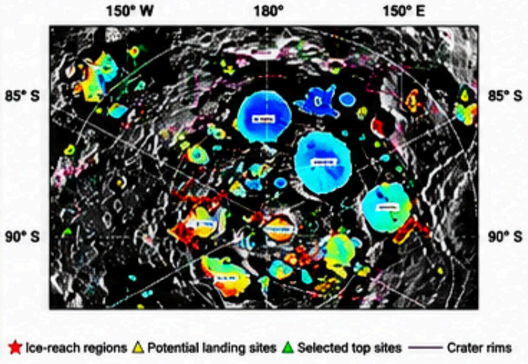












AIESA
ARTIFICIAL INTELLIGENCE & DATA SCIENCE ENGINEERING STUDENT ASSOCIATION

TEAM SHAURYAM

AITHON 2.0

Objective 3: Optimized Rover Traverse Path

Shared Data Layer
OHRC imagery, LOLA DEM, DFSAR CPR/DOP map, VFSD

Landing site
DIC3D-A* phase 1
PPO DRL Phase 2
DSC
PSR

Phase 1 - DIC3D-A*
Global illuminated approach
(Landing site --> crater rim)

$$Cost_{heuristic} = \alpha Cost_r + \beta Cost_D + \gamma Cost_{sl} + \delta Cost_{ice}$$

3D state space (x, y, t) search
Wait-mechanism at rim charging waypoints

Slip safety model

$$slip_{safe} = A\, e^{B\theta_l}$$

$$slip_{unsafe} = C\, e^{D\phi_r}$$

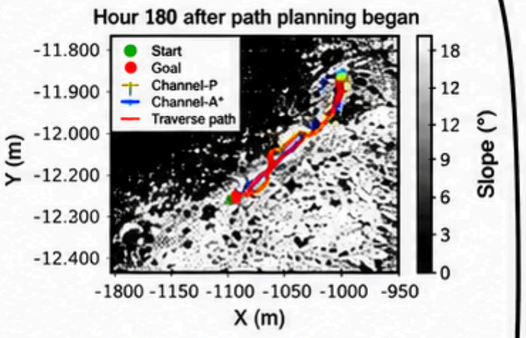
Phase 2 - PPO DRL
Battery-Gated DSC descent
(crater-rim --> DSC floor)

State Space
DEM Patch
CPR / DOP values
Battery state
Distance to ice-patch

Reward Function

$$r_t = r_t^{dis} + \alpha_s\, r_t^{safe}$$

Simulation Environment
Gazebo / ROS



Height, Slope, Roughness, Slip COSTs

Terrain Cost
Distance Cost

$$g(n) = \sum_{i=1}^{n} (\alpha Cost_r + \beta Cost_D + \gamma Cost_{sl} + \delta Cost_{ice})$$

$$h(n) = Distance(n, Goal)$$

$$f(n) = g(n) + h(n)$$

Illumination Cost
Ice Probability Cost
DFSAR CPR / DOP

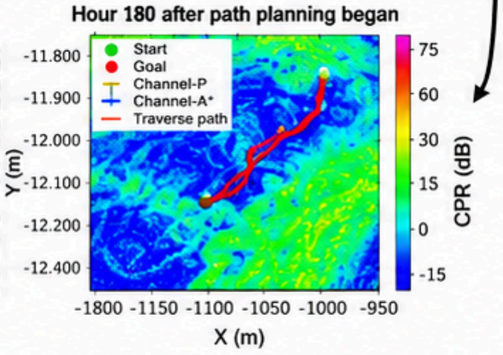
Phase Handoff Gate
(crater-rim)

Condition 1: Battery >= threshold
Condition 3: in experiment
Condition 2: VFSD < 0.3

All 3 Conditions met ?
F → Wait at rim
T

$$r_t^{dis} = \begin{cases} 10, & if\ (d_t^c < d_{goal}) \\ -10, & if\ (d_t^c > d_{max}) \\ d_t^{l-1} - d_t^l, & \text{otherwise} \end{cases}$$

$$r_t^{safe} = \begin{cases} -10, & (\theta_t > \theta_{max}) \text{ or } (\phi_t > \phi_{max}) \\ -(slip\ \beta_t), & \text{otherwise} \end{cases}$$



AIESA
ARTIFICIAL INTELLIGENCE & DATA SCIENCE ENGINEERING STUDENT ASSOCIATION

TEAM SHAURYAM

AITHON 2.0

Objective 4: Subsurface Ice Volume & Mass

1) Input Data

DFSAR L-band SRI
σ^0_{HH}, σ^0_{HV}, σ^0_{VV}
calibrated, map-projected

LOLA DEM
slope, aspect per pixel
local incidence angle

Ice detection mask
CPR > 1 & DOP < 0.13

2) Data Preprocessing

σ^0 calibration
$\sigma^0_{dB} = 20\log_{10}(DN) - k$

Speckle filter
Lee / Refined Lee filter

Local θ from DEM
pixel-wise incidence angle

3) Apply Ice-mask

Retain only CPR > 1 and DOP < 0.13 pixels
Removes rocky terrain false positives | Reduces ANN inference cost

4) IEM forward simulation

IEM parametrisation (from lunar in-setu obs)

$\varepsilon \in [1, 12]$ $\quad$ $RMS\ Height < 5$ cm (Gaussian) $\quad$ $RMS\ Height < 7.2$ cm (Exponential) $\quad$ $L = 24$ cm $\quad$ $\mu = 1$

2D Gaussian
$\exp\left(-\left(\frac{r}{L}\right)^2\right)$
smooth surface

2D Exponential
$\exp\left(-\frac{r}{L}\right)$
natural surfaces

3D Gaussian
2D → 3D extended isotropic surface

3D Exponential
3D + exp ACF
most realistic

Simulated training dataset (56K - 79K samples per model)
each sample: $(\sigma^0_{HH}, \sigma^0_{VV}, \theta_{local}) \rightarrow c_{theory}$ $\quad$ split 70 % train & 30 % test

5) ANN training

ANN Architecture(all 4 models)
Input (3): $(\sigma^0_{HH}, \sigma^0_{VV}, \theta_{local})$
Hidden: 3×64 Neurons
Output (1): $\hat{\varepsilon}$
Activation – tanh
Optimizer – Adam
Loss – MSE
Epochs – 50-150

Training performance
RMSE, MSE, R^2
on 30% test set

Validation (Apollo 17 proxy)
LRO Mini-RF S-band
best: 2DGauss R^2 = 0.82

6) Dielectric constant retrieval

Feed DFSAR into all 4 trained ANNs
Inputs: σ^0_{HH}, σ^0_{VV}, θ_{local} (per pixel)

4 parallel predictions -> (4 ε maps)
$\varepsilon^{Gauss}_{pred}$ ε^{Exp}_{pred} $\varepsilon^{Gauss}_{pred}$ ε^{Exp}_{pred}
discard pixels where $\varepsilon < 1$ or $\varepsilon > 12$

Weighted ensemble
$$\varepsilon_{final} = \frac{\sum_{i=1}^{4} R_i^2 \varepsilon_i}{\sum_{i=1}^{4} R_i^2}$$
$uncertainty = std(\varepsilon_1, \varepsilon_2, \varepsilon_3, \varepsilon_4)$
per pixels, propagated to final volume error bar

7) Looyenga-Landau-Lifshitz (LLL) mixing

$$\varepsilon_{eff}^{1/3} = v_{reg}\,\varepsilon_{reg}^{1/3} + v_{ice}\,\varepsilon_{ice}^{1/3} + v_{pore}\,\varepsilon_{pore}^{1/3}$$

$\varepsilon_{reg} \approx 2.7$ (Apollo regolith)
$\varepsilon_{ice} \approx 3.15$ (Water ice, 100 K)
$\varepsilon_{pore} = 1.0$ (Vacuum/Pore)

Solve for v_{ice} per pixel | Result: $f_{ice} \in [0, 1]$

8) Ice volume calculation (3 Scenarios)

Conservative
depth = 2m
porosity ϕ = 0.45
lower bound

Nominal
depth = 5m
porosity ϕ = 0.40
best estimate

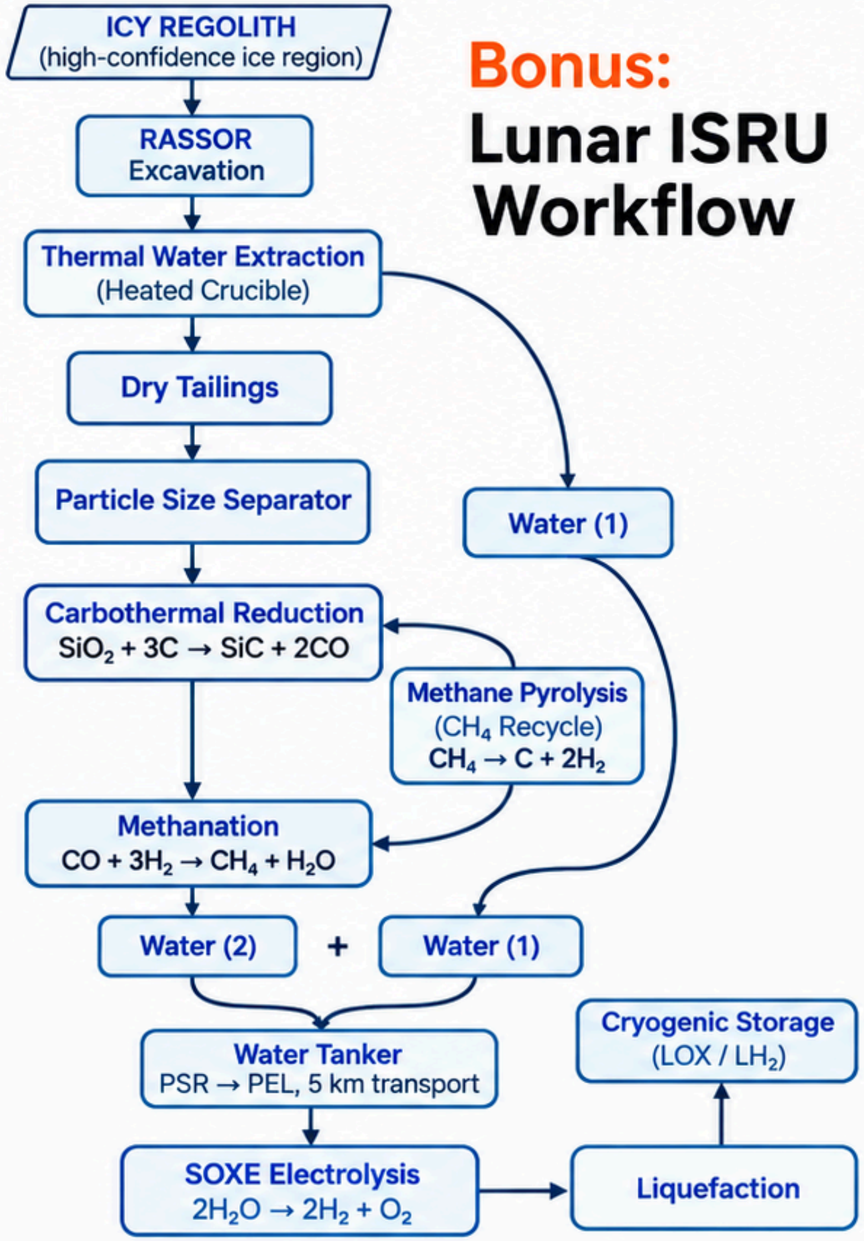
Optimistic
depth = 5m
porosity ϕ = 0.35
upper bound

$$V_{ice} = \sum_{pixels} (pixel_{area} \times depth \times f_{ice})$$

$V_{low}(m^3)$ $\quad$ $V_{nominal}(m^3)$ $\quad$ $V_{high}(m^3)$

$Mass_{ice} = V_{ice} \times 917\ kg\,m^{-3}$ $\quad$ **Final output (Vol_{ice}) = X (± Y%)**

RESEARCH AND REFERENCES



Bonus: Lunar ISRU Workflow

Tech Stack (Used in the Solution)

Programming & Data-Preprocessing: Python, NumPy, SciPy, Pandas, Rasterio, GDAL, GeoPandas
Radar & Image Processing: MIDAS, ENVI, OpenCV, Scikit-image
Machine Learning & AI: Gaussian KDE, ANN, PPO (Proximal Policy Optimization), Entropy-weighted FCM
Geospatial Analysis: LOLA DEM, Terrain Analysis, Illumination (VFSD) Modeling
Path Planning: DIC3D-A*, PPO-DRL, Slip-Safety Modeling
Dielectric & Ice Estimation: IEM, Particle Anisotropy, LLL Mixing Model, R^2-weighted Ensemble Learning
Visualization & Outputs: Matplotlib / Plotly, GeoTIFF, QGIS/ArcGIS
Simulation: Gazebo / ROS
Dashboard: Streamlit / React
Backend: FastAPI

References (Research Papers)

- ❑ **Sinha et al. (2026)** – *Subsurface Ice in Doubly Shadowed Craters as Revealed by Chandrayaan-2 DFSAR*
- ❑ **Jia et al. (2022)** – *Selection of Lunar South Pole Landing Site Using Fuzzy Cognitive Maps*
- ❑ **Miao et al. (2026)** – *Dynamic Illumination Constrained Spatio-Temporal Algorithm for Path Planning**
- ❑ **Sharma et al. (2023)** – *Integral Equation Modeling for Dielectric Retrieval Using Chandrayaan-2 DFSAR*
- ❑ **Ikeya et al. (2025)** – *Hybrid Lunar ISRU Plant: A Comparative Analysis with Carbothermal Reduction and Water Extraction*
- ❑ **Bhiravarasu et al. (2021)** – *Chandrayaan-2 DFSAR: Performance Characterization and Initial Results*
- ❑ **Campbell et al. (2012)** – *High Circular Polarization Ratios in Radar Scattering from Geologic Targets*
- ❑ **Raney et al. (2012)** – *The m-chi Decomposition of Hybrid Dual-Polarimetric Radar Data*
- ❑ **Yu et al. (2021)** – *Learning-Based End-to-End Path Planning for Lunar Rovers with Safety Constraints*
- ❑ **Putrevu et al. (2023)** – *Lunar Impact Craters: New Perspectives from Full-Polarimetric Analysis of Chandrayaan-2 DFSAR Data*
- ❑ **O'Brien (2022)** – *Double Shadows at the Lunar Poles*

"We are really passionate about advancing lunar exploration and are eager to contribute our ideas and technical expertise toward ISRO's future space missions."